

Applied Computer Vision for Engineering Systems

Syllabus design

This course provides a comprehensive, application-driven introduction to computer vision and deep learning. Students will progress from foundational optimization and neural networks to advanced modern paradigms, including generative models, self-supervised foundation models, and 3D neural rendering. The curriculum emphasizes cross-disciplinary deployment to solve high-stakes engineering and scientific challenges. Students will gain hands-on experience applying vision algorithms to real-world datasets across *field robotics*, *satellite remote sensing*, *climate tracking*, *extreme mining environments*, *energy infrastructure inspection*, and *structural bio-AI*. The course concludes with a collaborative, research-focused final project presentation.

Prerequisites: CSCI507 or CSCI470 or CSCI576 or EENG507.

Table 1: 16-Week Course Schedule

Module	Week Dates	Lecture Topic	Core Technical Focus & Key Algorithms
Module 1: CV & DL Basics	W1 8/25, 8/27	Introduction	Introduction and Course Overview
	W2 9/1, 9/3	Basics of Optimization	Brief history; Linear Classifier; Regularization
	W3 9/8, 9/10	CNN & Backpropagation	Convolutional neural networks (CNNs); Backpropagation; SGD
	W4 9/15, 9/17	RNNs & Attention & Transformers	Attention mechanisms; Transformers & Vision Transformers
Module 2: Advances in CV	W5 9/22, 9/24	Recognition & Localization	Classification, Detection and Segmentation (ResNet, YOLO, Fast R-CNN, Mask R-CNN, ViT, DETR)
	W6 9/29, 10/1	Video Understanding	Motion estimation with optical flow, temporal modeling with 3D CNNs, RNNs, and Video Vision Transformers;
	W7 10/6, 10/8	Large-Scale Distributed Training	GPUs & Cluster, Data/Context/Tensor/Pipeline Parallelism, How to train on GPUs
	W8 10/13 , 10/15	Self-supervised Learning	AutoEncoder (AE), Variational AE, Masked AE, Contrastive learning, CLIP, DINO
	W9 10/20, 10/22	Foundation Models	Large Language Models (LLMs), Vision Language Models (VLMs)
	W10 10/27, 10/29	Generative Models	Text-to-image and text-to-video generation with diffusion models
	W11 11/3, 11/5	Geometry & 3D Vision	3D Representations, Modern 3D Reconstruction & Generation Models (NeRF, 3DGS, VGGT, TRELLIS, Marble); Spatial Intelligence
Module 3: CV for Engineering Systems	W12 11/10, 11/12	Robotics & Autonomous	Robotic perception loops; VLA, navigation, embodied reasoning, motion planning & diffusion policy
	W13 11/17, 11/19	Remote Sensing & Earth Observation	Satellite imagery; multispectral/hyperspectral; geospatial segmentation; continuous temporal monitoring
	W14 11/24, 11/26	Mining & Energy	Industrial anomaly detection: wind turbine cracks, PV panel, gas / oil / power lines
	W15 12/1, 12/3	Climate & Environment	Disaster perception (flood, landslide, glacier, wildfire); Weather perception; Urban scene reconstruction
	W16 12/8	Healthcare	Medical image analysis & diagnosis (X-ray, MRI, pathology)
Final Exam	W17 12/02?	Project Presentation & Poster	Project Roadshow: 12 students (4 or 6 groups) presenting application-focused projects built on real-world datasets / their research